



MSME Idea Hackathon 6.0

Smart and Sustainable MSMEs

AeroPure :Transforming Vehicle Exhaust into Clean Air through Dynamic Flow Management and Nano-Scale Molecular Capture

Theme: Automotive technology

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1. Problem

Problem statement :

1. Pollution is not a problem from a single vehicle but collective pollution from all vehicles is a major problem.
2. In a moving highway, we will not identify any pollution, it isn't the same in traffic signals, we will feel how the air is polluted, it doesn't mean the highway is not polluted, that is why, taking in the account of collective pollution is necessary.
3. Even though, the new vehicles designed with less pollution concepts, in countries like India, it's obvious that many are using same vehicles for so many years, they find it difficult to comply with modern concepts.
4. Due to these pollution, government has to spend a lot in terms of :
 - 4.1. Health management
 - 4.2. Environmental management
 - 4.3. Deforestation

Existing Solution:

1. Instead of preventing, current solutions are aftermath solutions only, for example: Treatment is provided after getting affected by air pollution.
2. Conventional Static Diesel Particulate Filters : Standard factory-fitted or retrofit ceramic honeycomb filters used to capture soot.
3. Selective Catalytic Reduction Systems: Fluid-based injection systems coupled with catalysts to primarily target nitrogen oxides.

it will take more time and money to replace with existing technologies

what we are suggesting is a solution that will not take money and time .

we aren't modifying any engines, we are creating a portable filter fitted outside of the engine.

2. Customer Segments

1. **All Vehicle Owners:** Commercial fleet operators, logistics companies, and everyday diesel vehicle owners face constant struggles with strict urban emission norms, frequent DPF clogging during stop-and-go traffic, and expensive OEM replacement cycles. AeroPure addresses this by offering a cost-effective, modular aftermarket solution featuring quick-disconnect couplings and real-time dashboard monitoring that allows for rapid, hazard-free servicing and minimized vehicle downtime.

2. **Small & Medium Enterprises (MSME) Industries:** Small-scale manufacturing units, commercial complexes, and workshops relying heavily on diesel backup generators or localized industrial exhaust equipment require affordable ways to comply with pollution control board regulations. AeroPure provides these MSMEs with a compact, low-backpressure, multi-stage filtration system that delivers efficient particulate scrubbing and thermal management without the massive footprint or water consumption of traditional industrial wet scrubbers.

3. Unique Value Proposition (USP)

Key Benefits

- 1.Preventing Pollution at a Very Low Cost:** Designed with affordable, commercially off-the-shelf (COTS) components and replaceable modular layers.
- 2.Minimized Engine Strain & Low Backpressure:** Unlike conventional static filters that trap soot and choke the engine, AeroPure's integrated thermal management (heat sinks and fluid cooling) and graded pore structure maintain smooth exhaust flow to protect engine efficiency and save fuel.
- 3.Rapid, Hazard-Free Modular Servicing:**Equipped with self-sealing docking mechanisms and quick-disconnect couplings, the system allows operators or technicians to swap filter modules in minutes without toxic exposure or complicated teardowns.

Differentiators:

- 1.Effortless Government Norms & Compliance:** Built to proactively meet strict urban Pollution Control Board (PCB) and periodic vehicle emission test (PUC) standards, helping both vehicle owners and MSME generators avoid hefty fines and legal shutdowns.
- 2.Multi-Stage Active Thermal & Nano-Filtration:** Combines oil-cooled heat sinks, nano-engineered charcoal matrices, and synthetic arrays into a single compact unit outperforming single-stage DPFs that frequently fail under heavy thermal stress.
- 3.Real-Time Digital Monitoring:** Unlike traditional passive exhaust pipes that operate blindly, AeroPure integrates live sensor tracking and dashboard alerts to notify users of filter saturation before performance drops.

Tagline:

Modular Filtration, Zero Compromise: Smart Exhaust Purification for Vehicles and MSME.

4. Competitive Advantage

How your product is advantageous than your competitor's.

Compare the advantages in a tabular form.

Feature	Your Product	Competitor
Size of product	Small and compact	Larger
Cost of product	Affordable and low	higher
Running cost	Very low cost	Very high
Lifetime	Longer years(approx-10 years)	Not applicable

5. Solution

Product Overview:

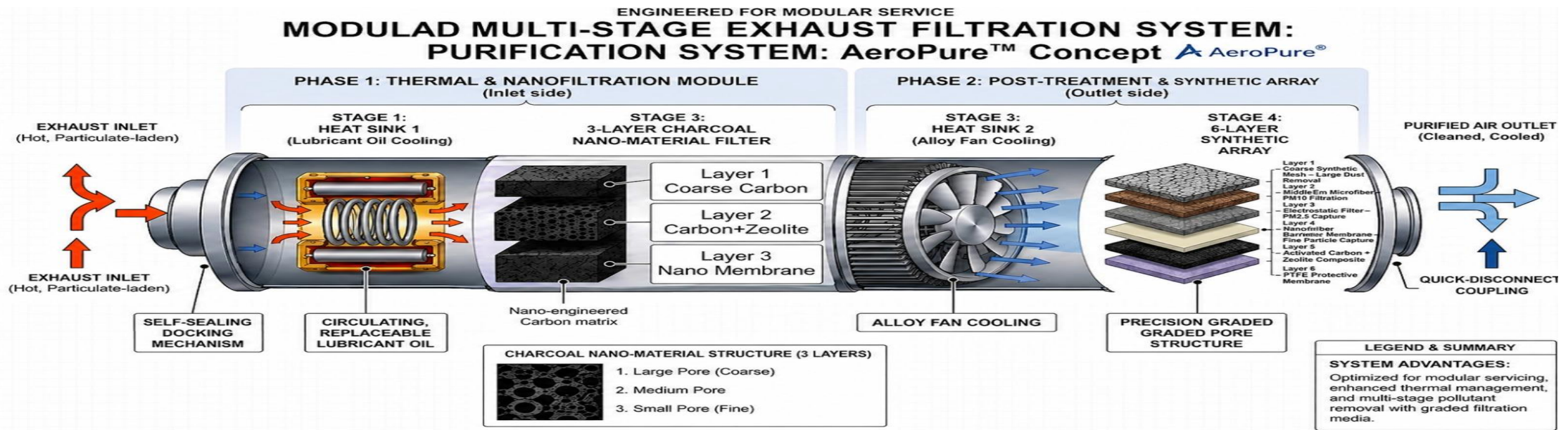
- 1.Multi-Stage Membrane & Nano-Filtration:** Combines oil-cooled heat sinks, nano-engineered charcoal matrices, and graded synthetic arrays into a compact unit to capture fine particulate matter effectively.
- 2.Real-Time Sensor Diagnostics:** Integrates continuous particulate and temperature monitoring to track filter saturation levels and alert users before efficiency drops.
- 3.Modular Service Architecture:** Built with quick-disconnect couplings and self-sealing docking mechanisms that allow individual filter layers to be swapped out rapidly and safely.

Problem–Solution Fit:

- 1.Easily Fitted & Portable:** Designed as a compact, retrofit unit that can be easily installed on diverse vehicle exhaust tailpipes and small MSME generator stacks without altering core engine architecture.
- 2.Eliminates Urban Clogging & Backpressure:** Solves the frequent urban stop-and-go clogging issue of conventional DPFs by utilizing optimized thermal management and low-backpressure graded pores to protect engine performance.
- 3.Affordable Maintenance vs. Expensive Replacements:** Replaces costly OEM filter overhauls with inexpensive, modular replacement layers that fit a student-friendly or small-business budget.

SDG: 3,9,11

TRL: 3



Phase 1, Stage 3: 3-Layer Charcoal Nano-Material Filter

- **Layer 1 (Coarse Carbon):** Features a large pore network to capture macro-particles and heavy volatile compounds early in the exhaust stream.
- **Layer 2 (Carbon + Zeolite):** Utilizes medium pores to combine organic gas adsorption with molecular sieving for medium-sized molecules and gas-phase contaminants.
- **Layer 3 (Nano Membrane):** Contains small, fine pores acting as a high-density barrier against microscopic particulates and complex chemical structures.

Phase 2, Stage 4: 6-Layer Synthetic Array

- **Layer 1 (Coarse Synthetic Mesh):** Acts as a mechanical pre-filter for large dust removal to protect subsequent finer layers.
- **Layer 2 (MiddleEm Microfiber):** Engineered for PM10 filtration to capture dust, pollen, and larger mold spores.
- **Layer 3 (Electrostatic Filter):** Employs a static electric charge to attract and trap dangerous PM2.5 combustion particles and fine organic compounds.
- **Layer 4 (Nanofiber Barrier):** Features a tight membrane layer dedicated to stopping sub-micron pollutants through mechanical interception.
- **Layer 5 (Activated Carbon + Zeolite Composite):** Provides secondary gas-phase purification to adsorb remaining odors, chemical vapors, and toxic gasses.
- **Layer 6 (PTFE Protective Membrane):** A durable, chemical-resistant Polytetrafluoroethylene barrier protecting the final filter array integrity before clean air exits.

6. Channels

Customer Acquisition & Distribution Channels

1. Automobile Service Centers & Aftermarket Stores: Partnering with local garages, vehicle service stations, and automotive spare parts shops to stock and recommend AeroPure as a retrofittable upgrade during routine vehicle maintenance and PUC checks.

2. Hardware & Industrial Equipment Suppliers: Collaborating with local hardware and generator dealers who cater to Small and Medium Enterprises (MSMEs), allowing businesses to easily purchase exhaust filtration add-ons alongside backup power equipment.

3. Automotive Repair Workshops & Fabricators: Engaging directly with local mechanic workshops and metal fabrication hubs to install, service, and swap modular filter cartridges using quick-disconnect couplings.

4. Direct Sales & B2B Fleets: Leveraging direct outreach to commercial logistics companies, delivery fleets, and small-scale industrial unit owners to offer customized bulk installations and pilot demonstrations.

7. Revenue Streams

- 1. Direct Hardware Unit Sales:** One-time retail sales of complete AeroPure filtration systems to individual vehicle owners, logistics fleet operators, and Small and Medium Enterprise (MSME) diesel generator owners.
- 2. Modular Filter Cartridge Replacements (Recurring Revenue):** Selling affordable, consumable replacement layers (such as nano-engineered charcoal matrices and graded synthetic filter arrays) as periodic maintenance accessories when filters saturate.
- 3. B2B Fleet Maintenance Subscriptions:** Offering scheduled maintenance contracts and bulk replacement packages to commercial delivery fleets, bus operators, and industrial facilities.
- 4. Value-Added IoT Dashboard / Diagnostics Upgrades:** Providing premium digital features, advanced emission-tracking analytics, and automated maintenance alert subscriptions for commercial fleet managers.

8. Cost Structure

Phase 1: Thermal & Nanofiltration Module (Inlet Side)

Stage 1: Heat Sink & Lubricant Cooling Module*

Cost Range: ₹1,200 – ₹1,800 per unit

Low-Cost Optimization: Utilizing standard stamped aluminum cooling fins and recycled-grade metal tubing instead of expensive exotic high-temperature metal alloys.

Stage 3: 3-Layer Charcoal Filter (Budget Configuration)

Cost Range: ₹800 – ₹1,200 per unit

Breakdown:

Layer 1 (Coarse Carbon): Standard commercial granular activated carbon (₹150 – ₹250).

Layer 2 (Carbon + Zeolite Blend): Bulk-mixed industrial zeolite and low-cost carbon powder (₹300 – ₹450).

Layer 3 (Fine Mesh Barrier): Standard multi-weave fine metal wire mesh replacing costly proprietary nano-membranes (₹350 – ₹500).

Phase 2: Post-Treatment & Synthetic Array (Outlet Side)

Stage 3 (Outlet): Secondary Cooling & Fan

Cost Range: ₹900 – ₹1,400 per unit

Low-Cost Optimization: Standard off-the-shelf exhaust-rated cooling fans and simplified heat dissipation plates.

Stage 4: 6-Layer Synthetic Array (Simplified Cost-Effective Media)

Cost Range: ₹1,500 – ₹2,300 per unit

Breakdown:

Layer 1 (Coarse Synthetic Mesh): Standard polypropylene mesh pre-filter (₹100 – ₹150).

Layer 2 (Microfiber Layer): Non-woven polyester fiber for dust and PM10 (₹150 – ₹250).

Layer 3 (Electrostatic Layer): Standard electret synthetic needle-felt (₹250 – ₹350).

Layer 4 (Fine Particulate Barrier): Multi-layer cellulose-synthetic blended paper media (₹300 – ₹450).

Layer 5 (Adsorption Layer): Low-density carbon-impregnated foam mat (₹400 – ₹600).

Layer 6 (Protective Backing): Standard high-density polyethylene (HDPE) or economic mesh backing instead of expensive PTFE membranes (₹300 – ₹500).

External Hardware & Casing

1. Docking Mechanism & Couplings: ₹700 – ₹1,100 per unit

2. Utilizing standard industrial quick-release snap fittings and molded rubber seals instead of custom-machined precision valves.

3. Casing & Assembly Overheads: ₹600 – ₹900 per unit

4. Standard sheet-metal rolling for the outer sleeve and automated assembly to reduce labor costs.

Summary Cost Profile (Low-Cost Production)

1. Total Estimated Direct Production Cost (COGS): ₹5,700 – ₹8,700 per unit

9. Key Metrics

Business KPIs

Customer Metrics:

1. Retention & Replacement Rate: Tracking how frequently customers return to purchase consumable filter cartridge replacements.
2. Emission Compliance Success: Measuring the percentage of vehicle and MSME generator owners who successfully pass local pollution control board tests.

Financial Metrics:

1. Gross Profit Margin: Calculating the profitability per unit based on low-cost raw materials and COTS components.
2. Recurring Revenue Share: Monitoring the steady income generated specifically from periodic modular filter replacements and maintenance subscriptions.

10. Unfair Advantage

Unfair Advantage & Competitive Moat

Unique Strategic Assets:

1. Proprietary Modular Architecture: The self-sealing docking mechanism and quick-disconnect coupling design allow for instant, tool-free layer swaps, completely bypassing the welded, non-serviceable limitations of conventional exhaust filters.

2. Integrated Dual-Phase Thermal & Nano-Filtration: Combining oil-cooled heat sinks with a multi-layered nano-charcoal and graded synthetic array into a single compact unit delivers superior particulate capture without high backpressure.

Long-Term Defensibility:

1. Low-Cost Scalability: Built using affordable Commercial Off-The-Shelf (COTS) materials, creating high entry barriers for competitors trying to match our low production and replacement costs.

2. High Switching Costs & Recurring Ecosystem: Once vehicle fleets and MSME operators adopt the modular housing, they remain locked into purchasing proprietary, low-cost consumable filter cartridge replacements.

THANKYOU